

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Digital Logic (New)

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Signal and differentiate between the Digital and Analog signal. 7
- b) Perform the following operations 8
 - i. $(E8E)_{16} = ()_8$
 - ii. $(349)_{10} = ()_{BCD}$
 - iii. $(55)_{10} - (67)_{10}$ Using 2's Complement
 - iv. $(1001101)_2 = ()_{gray}$
2. a) Why NAND and NOR gates are called Universal gate? Draw and explain about all the basic gates that can be derived from NOR gate. 8
- b) Draw K-map in SOP form for expression $F(A,B,C,D) = \sum(2,3,8,11,13,15)$ and $D = \sum(1,7,10,14)$ Also realize using NAND Gates only. 7
3. a) Design Full Adder using two half adders and an OR gate. 7

OR

Design 3-bit Odd parity generator and Checker.

- b) Design a combinational logic circuit that a voting system has four inputs A, B, C & D. If any three or all four inputs are HIGH the output must go HIGH also, If the voting is 2/2 then the output must take up the state of input D. If only one, or none of the inputs is HIGH the output must be LOW. 8
4. a) Design 2-bit Magnitude Comparator. 7
- b) Implement the following Boolean function 8
$$F(A, B, C, D) = \sum(0, 1, 2, 6, 11, 13, 14)$$
 using 8 to 1 Multiplexer and B,C,D as selection lines.

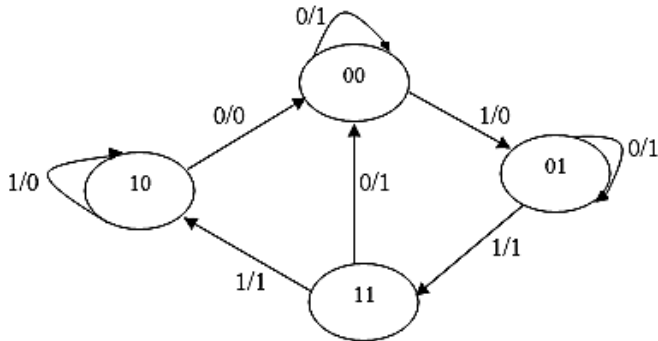
OR

A combinational circuit is defined by the function

$$F_1(X, Y) = XY + XZ' \text{ and } F_2(X, Y) = XZ' + YZ$$

Implement the circuit with a PLA having three inputs, three product terms and two outputs.

5. a) What is the drawback of S-R Flip-flop? Explain the operation of J-K Flip-flop with its logic diagram, truth table and excitation table. 8
- b) Design Sequential circuit using JK flip flop from the following state diagram. 7



6. a) Design 3-bit UP/DOWN Counter. 8
- b) Design a 4-bit arithmetic circuits unit that performs eight different arithmetic operations and logical unit with four different operations. 7
7. Write short notes on: (**Any two**) 2×5
 - a) Weighted VS Non-Weighted Codes
 - b) Master Slave Flip-flops
 - c) Output hazard race